Calculations report: TEST INPUT - readiness trace pump panel

Job number: TEST-001

Revision A, 2026-10-23

NOT APPROVED: for review only, not for execution.

The design rules are this software's defaults; the company has not confirmed them.

Basis of calculation

- Design current Ib = P / (k U cos phi); breaker In >= Ib, cable Iz >= In (IEC 60364-4-43 §433.1; ABB Electrical Installation Handbook Vol. 2, §2.1-2.3).
- Current-carrying capacity from IEC 60364-5-52 Annex B, corrected for ambient temperature and grouping.
- Voltage drop from the origin, from Schneider EIG Fig. G28 (copper) and ABB §2.2.2 (aluminium), against IEC 60364-5-52 Annex G or the company's limits.
- Fault level at a sub-board through its feeder, IEC 60909-0 (cmax 1.10); breaking capacity at or above it.
- Earth fault disconnection, IEC 60364-4-41 §411.4.4: Zs x Ia <= 0.95 U0, Ia the top of the breaker's instantaneous band (IEC 60898-1).
- Short circuit along the cable, ABB §2.4: the far-end Ikmin trips the breaker at once; k^2S^2 from Table 1 for comparison with the breaker's let-through I^2t.

Board PP1

400 V, 3-phase, TN-S · Ik 10 kA · Ze - ohm · Incomer C100 4P 10 kA

Circuit	Ib A	Protection	Cable	L m	dU %	Zs ohm	Zs max	k^2S^2 $(kA)^2s$
Panel lighting (test)	1	C200 1P 10 kA	3G1.5 Cu PVC	10	0.12	-	0.11	0.03
Duty pump (test)	15.2	MA20 3P	4G1.5 Cu PVC	30	2.85	-	-	0.03
Standby pump (test)	56	MA80 3P	7G6 Cu PVC	30	1.82	-	-	0.476
Booster (test)	10.4	25 A	4G1.5 Cu PVC	20	1.3	-	-	0.03

Notes

- Group 1 breaker raised to 16 A, 1.6 x the 10 A breaker after it, for discrimination.
- Duty pump (test): Ir 15.2 A, the typical Ir of a 7.5 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate.
- Duty pump (test): dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2.
- Standby pump (test): Ir 56 A, the typical Ir of a 30 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate.
- Standby pump (test): star_delta starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 5: 400 V 50 kA Y/D Normal Type 2.
- Standby pump (test): star and delta contactors need a mechanical interlock; the motor is fed by six conductors (U1 V1 W1, U2 V2 W2).

- Booster (test): Ir 10.38 A from 5.5 kW shaft power at cos phi 0.85, efficiency 0.9; check it against the nameplate.
- Booster (test): ACS880-01-12A6-3 for normal duty at 30 °C, behind 25 A aR fuses (ABB ACS880-01 hardware manual (3AUA0000078093)). Use a screened, symmetrical motor cable, earthed 360° at both ends.
- Booster (test): the fuses need at least 75 A prospective short-circuit current at the board to operate fast enough.
- Breakers are rated for at least 10 kA breaking capacity against 10 kA prospective at the board; each RCCB's conditional short-circuit current with the breaker ahead of it must reach it too.
- Leave 1 spare outgoing ways (20 %).
- Loads without a power factor were taken at cos phi 0.9.
- Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.
- No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze.
- Each cable's withstand k^2S^2 is in the cable list; compare it with the selected breaker's let-through I^2t at 10 kA from its maker's curve (ABB handbook §2.4).
- Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.
- Circuit rules are this software's defaults, not confirmed by the company's engineers.
- Enclosure: none named, since 11 devices have no width (rail_widths_mm in the company settings).

Findings (1 blocking, 9 warning, 10 info)

[BLOCKING] PP1 / c1: Panel lighting (test): 200 A protection is above the 17.5 A its 3G1.5 cable carries (method B1); In <= Iz does not hold. (integrity_device_above_cable_capacity; source: ABB Electrical installation handbook Vol. 2, §2.3 (In <= Iz); Iz from §2.2.1 Tables 4, 5, 8)

[WARNING] PP1 / c2: Duty pump (test): Ir 15.2 A, the typical Ir of a 7.5 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate. (motor_current_table; source:

policy; engineering confirmation required)

[WARNING] PP1 / c3: Standby pump (test): Ir 56 A, the typical Ir of a 30 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate. (motor_current_table; source:

policy; engineering confirmation required)

[WARNING] PP1 / c4: Booster (test): Ir 10.38 A from 5.5 kW shaft power at cos phi 0.85, efficiency 0.9; check it against the nameplate. (motor_current_formula; source: policy; engineering confirmation required)

[WARNING] PP1 / c4: Booster (test): the fuses need at least 75 A prospective short-circuit current at the board to operate fast enough. (drive_fuse_fault_level; source: policy; engineering confirmation required)

[WARNING] PP1: Loads without a power factor were taken at cos phi 0.9. (default_power_factor; source: policy; engineering confirmation required)

[WARNING] PP1: No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze. (earth_fault_no_ze; source: policy;

engineering confirmation required)

[WARNING] PP1: Each cable's withstand k^2S^2 is in the cable list; compare it with the selected breaker's let-through I^2t at 10 kA from its maker's curve (ABB handbook §2.4). (short_circuit_withstand; source: policy;

engineering confirmation required)

[WARNING] PP1: Circuit rules are this software's defaults, not confirmed by the company's engineers. (rules_unconfirmed; source: policy; engineering confirmation required)

[WARNING] PP1: Enclosure: none named, since 11 devices have no width (rail_widths_mm in the company settings). (enclosure_unknown_widths; source: policy; engineering confirmation required)

[INFO] PP1 / c4: Booster (test): the cable after a drive is protected by the drive's motor overload setting, which is not checked here. (integrity_drive_cable_unchecked; source: policy)

[INFO] PP1: Group 1 breaker raised to 16 A, 1.6 x the 10 A breaker after it, for discrimination. (discrimination_group_raised; source: policy)

[INFO] PP1 / c2: Duty pump (test): dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2. (starter_table; source: policy)

[INFO] PP1 / c3: Standby pump (test): star_delta starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 5: 400 V 50 kA Y/D Normal Type 2. (starter_table; source: policy)

[INFO] PP1 / c3: Standby pump (test): star and delta contactors need a mechanical interlock; the motor is fed by six conductors (U1 V1 W1, U2 V2 W2). (star_delta_interlock; source: policy)

[INFO] PP1 / c4: Booster (test): ACS880-01-12A6-3 for normal duty at 30 °C, behind 25 A aR fuses (ABB ACS880-01 hardware manual (3AUA0000078093)). Use a screened, symmetrical motor cable, earthed 360° at both ends.

(drive_selected; source: policy)

[INFO] PP1: Breakers are rated for at least 10 kA breaking capacity against 10 kA prospective at the board; each RCCB's conditional short-circuit current with the breaker ahead of it must reach it too.

(breaking_capacity; source: policy)

[INFO] PP1: Leave 1 spare outgoing ways (20 %). (spare_ways; source: policy)

[INFO] PP1: Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip

(about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables. (discrimination; source: policy)

[INFO] PP1: Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation. (cable_conditions; source: policy)

Prepared	Checked	Review approved
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